Credit Name: CSE 2110 Procedural Programming 1
Assignment Name: AddCoins.java
The following questions can help you in thinking critically about your problem-solving processes:
Understanding the Problem
How did you approach understanding the challenge?
I looked at how to isolate the first letter of a string using charAt(0) and how to compare character values alphabetically using relational operators.
Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?
Handling uppercase and lowercase input differences was tricky, but I resolved it by using Character.toUpperCase() so the conditional ranges only need to check capital letters.
Planning the Solution
Did you create a plan or break the problem into smaller steps before coding?
Yes, I broke it down into getting the name inputs, extracting and capitalizing the initial letter, evaluating the alphabetical ranges, and printing out the formatted result string.
How did you decide on the tools, data structures, or algorithms to use?
I used String variables for names, char for the initial letter, and an if-else if ladder to separate the alphabet into three distinct groups.
Implementation
Did you write the code in small pieces or attempt the entire solution at once?
I wrote it in pieces: first setting up the Scanner and string inputs, then writing the character extraction logic, and finally adding the conditional group ranges and output print statement.
How did you test your solution along the way to make sure it was working?
I tested names starting with letters from each range (like 'A', 'M', and 'Z') to ensure the correct group number was assigned for each category
Overcoming Challenges
What part of the problem was the most difficult for you?
Ensuring the program wouldn't crash if an empty string or unexpected character was entered for the last name
How did you handle moments when you felt stuck or unsure of what to do next?
I added a safety check (lastName.length() > 0) to make sure the program only evaluates characters if the string actually contains text
Learning
Was there anything you learned that you think will help you with future challenges?
I gained a lot of confidence using string methods like charAt() and toUpperCase(), which are super useful for text-processing programs